

SPECIFIC AIMS

Oropharyngeal cancer (OPC) caused by oncogenic human papillomavirus (HPV) is increasing rapidly in incidence in the general population of the United States and at an even more rapid pace among US Veterans. Although it is generally associated with favorable outcomes, OPC in the Veteran population is associated with more aggressive disease with lower progression free (PFS) and overall survival (OS). There are currently no extant estimates of HPV (+) OPC burden in the Veterans Health Administration (VHA) which makes it exceedingly difficult to develop prevention or treatment strategies aimed at Veterans. We do not believe that simply extrapolating general population data to the Veteran population is appropriate for 2 reasons. First, although HPV vaccination holds the promise of primary prevention of HPV-associated cancers, recent data suggest that vaccination rates in active-duty military and Veteran populations are extremely low. The Center for Disease Control recently expanded HPV vaccination indications to include men and women <45 years of age, making many Veterans entering the VA healthcare system eligible. Second, Veterans smoke at a higher rate compared to the general population, and it is also known that tobacco use diminishes prognosis in HPV (+) OPC.

The objective of this proposal is to understand the current and future burden of HPV (+) OPC in the VHA population by examining barriers to HPV vaccination, measuring disease incidence and outcomes in a national VHA HPV (+) OPC cohort, and developing a tool to model the potential benefit of primary and secondary prevention on HPV (+) OPC incidence and mortality. Completion of the aims detailed below should generate a clinical meaningful impact on individual Veterans with OPC and potentially generate more cost-effective approaches to this deadly disease within the VHA. The Specific Aims of this proposal are:

Aim 1. To measure the current burden of HPV+ OPC in the VHA population impact of tobacco exposure on HPV(+) OPC outcomes in a contemporary VHA cohort. Using a VHA OPC working group established at the 2019 OPC Field Based Meeting we will assemble and analyze a multi-institutional Veteran cohort of ~1,500 Veterans with HPV (+) OPC (2010-2018) to measure oncologic outcomes in the VHA and the the relative impact of smoking on progression free survival (PFS) and overall survival (OS). We hypothesize that heavy tobacco exposure (>20 pack-years) will generate a disproportionately lower PFS and OS in Veterans compared to non-Veterans with HPV (+) OPC.

Aim 2. Define the prevalence of HPV vaccination among eligible Veterans and identify populations at highest risk for under-vaccination: Using data from the VHA Corporate Data Warehouse (CDW), we will estimate the prevalence of HPV vaccination in Veterans <45 years of age and examine vaccination rates by race/ethnicity, socioeconomic status, smoking status, rurality based on Rural-Urban Commuting Area Codes, and geographical location based on Veteran Integrated Service Network (VISN). VISNs are responsible for VHA healthcare planning and resource allocation in a particular geographic region, and thus present an ideal opportunity to assess geographic variability in vaccination rates and ultimately apply targeted interventions. We hypothesize that smokers and African-Americans have the lowest rate of HPV-vaccination, and HPV vaccination rates will vary significantly across regions, with rural, South-central, and Southeast regions demonstrating the lowest rates of HPV vaccination. This approach will identify at-risk groups based on demographic or regional characteristics, and allow for tailored VHA or VISN-specific HPV vaccination campaigns or other cancer prevention strategies.

Aim 3: To project future HPV (+) OPC incidence and cancer-related mortality in the VHA. A Veteran-centric, evidence-based HPV (+) OPC simulation model will be constructed to describe disease progression in relation to risk factors, diagnosis, treatments, and survival. This model will incorporate data on current HPV vaccination and smoking rates in the VHA population and quantify disease burden in terms of HPV (+) OPC incidence, survival time, quality-adjusted life years, and cost-effectiveness. Additionally, we will model the potential impact of primary (HPV vaccination) and secondary (smoking cessation) on incidence and outcomes in HPV (+) OPC. We hypothesize that tailored HPV vaccination and smoking cessation efforts will reduce HPV (+) OPC incidence beginning in 2040, with a delayed improvement in survival and accompanying reduction in healthcare utilization and cost 3-5 years later.